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Spatial pattern modelling of wildfires in Catalonia, Spain 2004–2008

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ABSTRACT

The paper has three objectives: firstly, to evaluate how the extent of clustering in wildfires differs across the years they occurred; secondly, to analyse the influence of covariates on trends in the intensity of wildfire locations; and thirdly, to build maps of wildfire risks, by year and cause of ignition, in order to provide a tool for preventing and managing vulnerability levels. For these objectives we analysed the spatio-temporal patterns produced by wildfire incidences in Catalonia, located in the north-east of the Iberian Peninsula. The methodology used has allowed us to quantify and assess possible spatial relationships between the distribution of risk of ignition and causes. These results may be useful in fire management decision-making and planning. The methods shown in this paper may contribute to the prevention and management of wildfires, which are not random in space or time, as we have shown here.

1. Introduction

A wildfire is any uncontrolled fire in combustible vegetation that occurs in the countryside or a wilderness area. A wildfire differs from other fires in its extensive size, the speed at which it can spread out from its original source, its potential to change direction unexpectedly, and its ability to jump gaps such as roads, rivers and fire breaks (National Interagency Fire Center, 2011). Wildfires are characterized in terms of the cause of ignition, their physical properties such as speed of propagation, the combustible material present, and the effect of weather on the fire (Flannigan et al., 2006).

The four major natural causes of wildfire ignitions are lightning, volcanic eruption, sparks from rock falls, and spontaneous combustion (National Wildfire Coordinating Group, 1998; Scott, 2000). McRae (1992), among others, suggests that lightning ignitions do not occur anywhere, but favour locations satisfying certain terrain conditions. However, many wildfires are attributed to human sources (Pyne et al., 1996). First, there are human actions that directly cause ignitions deliberately or accidentally. In addition to this, great social upheaval in the last century has led much of the population to move from rural to urban areas. Many areas have witnessed an abandonment of farming and livestock practices, which leads to an accumulation of fuel for fires to feed on. The abandonment of rural areas has also reduced the capacity among the population for noticing fires and taking action when they first begin (William et al., 2000; Badia et al., 2002).

Fire risk is very high in the Mediterranean region due to its marked seasonality, with high temperatures and low humidity in summer, and these climatic trends interact with landscape dynamics. In the case of Catalonia, the process of afforestation in different agricultural areas and the increasing abandonment of rural activities have led to a situation of extreme vulnerability to fires, especially in Mediterranean mountain areas, where the aforementioned factors have led to forests being abandoned and their subsequent expansion, proliferation and interconnection (Loepfe et al., 2011). In addition to this, other factors, such as constructing second homes in these forest areas, the proliferation of roads and electricity networks and an increased flow of people make this region more susceptible to the ignition of large forest fires (Díaz-Delgado and Pons, 2001; Moreira et al., 2001).

Given that fire is a naturally occurring element in the Mediterranean ecosystem, the prevention and suppression of forest fires needs to be addressed so as to minimize risk and vulnerability of society.

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In fact, there are by now many studies of the spatial patterns of wildfire risk in various locations around the globe. Without being exhaustive, and referring only to more recent we cite works on fires, above all, in North America (Chen, 2007; Yang et al., 2008; Gedalof, 2011; Miranda et al., 2011; Gralewicz et al., 2011) but also in the Mediterranean region (Millington, 2005; Millington et al., 2009, 2010; Romero-Calcerrada et al., 2010), Asia (Liu et al., 2012) and Oceania (O'Donnell, 2011), among many others.

If we associate wildfires with their spatial coordinates, the longitude and latitude of the centroid of the burned area or the place where they were detected, along with other variables such as size or cause, and we know the time at which they started, it is possible to identify them by means of a spatial-temporal stochastic process. Such processes, called spatial point processes, often present dependences between the spatial positions and time instants, as well as interdependence between one another.

Spatial point processes are complex stochastic models that describe the location of events of interest and occasionally some information on these events. The most common models are those where the locations are given in two dimensions. Univariate point processes include only the location of events; point processes with marks (or marked point processes) include additional information about each event. These data sets can be used to respond to a variety of questions. The scientific context of these questions depends on the area of application, but they can be classified into three broad groups. First, one might be interested in whether the spatial pattern for the observed data is grouped, distributed regularly or random. A second group of questions would refer to the relationship between different types of events in a marked process or process with marks (variables measured only at fire locations, such as size of area burned, cause of fire or year of occurrence). The third and last group of questions focuses on density (number of events per unit area).

What is usually of more interest is to detect trends in the intensity of fire locations (i.e. probability of occurrence), and to determine how (or whether) such trends are influenced by covariates, observable at each location of the spatial window. These covariates might include vegetation or land use, other descriptors of terrain (such as elevation, slope and orientation), and others such as proximity to human populations or to concomitants of human activity (roads and railroads). Interaction between points may generally be of some interest in its own right. More important is the impact of the presence of interaction on statistical inference concerning trends and its dependence on covariates. Temporal clustering of wildfires, whether deriving from multiple ignition lightning events, arson (Butry and Prestemon, 2005), or other sources, combined with favourable fuel and weather conditions, can force suppression resource rationing across space. Spatial clustering can also indicate the presence of risk factors.

This paper has three objectives. Firstly, to evaluate how the extent of clustering in wildfires differs across marks, in particular years and causes of wildfire ignition. Secondly, to analyse the influence of covariates such as land use, slope, aspect and hill shade on trends in the intensity of wildfire locations. And finally, to build maps of wildfire risk, by year and cause of ignition, in order to provide a tool for managing vulnerability levels.

2. Methods

2.1. Data set

We have analysed the spatio-temporal patterns produced by wildfire incidences in Catalonia, located in the north-east of the Iberian Peninsula. The region is bordered by mountain landscapes, the Pyrenees in the north and the Iberian System in the south. The region is further delimited by the Ebro river to the south and south-west, and the Mediterranean coast to the east. It is a region with a surface area of 30,000 square kilometres (12,355 sq mi), representing 6.4% of the Spanish national territory.

The total number of fires recorded in the area studied during the period 2004–2008 was 3083. In addition to the locations of the fire centroids, measured in Cartesian coordinates (Mercator transversal projections, UTM, Datum ETRS89, zone 31-N), several marks and covariates were considered.

Variables measured only at fire locations are called marks. In this paper, marks included the year the wildfire occurred (from 2004 to 2008) and also the cause of ignition of each wildfire (classified as: natural causes; negligence and accidents; intentional or arson; and unknown causes and revived).

Spatial covariates were also considered. In particular, three continuous covariates: slope, aspect and hill shade; and one categorical variable: land use. Land use will obviously affect fire incidence, but, moreover topographic variables (slope, aspect and hill shade) affect not only fuel and their availability for combustion (Ordóñez et al., 2012) but also have effects on weather, inducing several local wind conditions including slope and valley winds. In fact, Dillon et al. (2011) point out that topographic variables were relatively more important predictors of severe fire occurrence than either climate or weather variables.

Slope is the steepness or degree of incline of a surface. In this paper, the slope for a particular location was computed as the maximum rate of change of elevation between that location and its surroundings. Slope was expressed in degrees. Aspect is the orientation of the slope, measured clockwise in degrees from 0 to 360, where 0 is north-facing, 90 is east-facing, 180 is south-facing, and 270 is west-facing. Hill shading is a technique used to visualize terrain as shaded relief, illuminating it with a hypothetical light source. Here, the illumination value for each raster cell was determined by its orientation to the light source, which, in turn, was based on slope and aspect. With respect to land use variables, we used the CORINE database (Coordination of Information on the Environment). The CORINE program was initiated in 1985 by the European Commission and was adopted by the European Environment Agency (EEA) in 1994. The main objective of the CORINE program is to capture numerical data and geographical order to create a European database on environment for certain priority topics such as land cover and biotopes (habitats), through the interpretation of images collected by the Landsat series of satellites and SPOT. Although this is based on remote sensing images as a data source it is actually a photo interpretation project and not automated classification. In this paper we have used the CORINE land cover map for the year 2006 (European Environment Agency, 2007). Data are gathered on a 1:100,000 scale with a minimum mapping unit (MMU) of 25 ha; the linear elements listed are those with a width of at least 100 m. The database includes 44 categories, in accordance with a standard European nomenclature, organized into five large groups: artificial surfaces, agricultural areas, forest and semi-natural areas, wetlands and water bodies (Heymann et al., 1994). In this paper we reclassified land use into ten categories: coniferous forests; dense forests; pastures; fruit trees and berries; artificial non-agricultural vegetated areas; transitional woodland scrub; scrub; natural grassland; mixed forests; and urban, beaches, sand, bare rocks, burnt areas and water bodies.

In order to model the dependence of a point pattern on a spatial covariate, there are several requirements. Firstly, the covariate must be a quantity observable at each location in the window (e.g. slope, aspect and hill shade). Such covariates may be continuous values or factors (e.g. land use). Secondly, the values of the covariate at each point of the data point pattern must be available. Thirdly, the values at some other points in the window must also be available.

2.2. Statistical methods

The simplest of all possible point process models is the constant intensity Poisson process, frequently referred to as the model of complete spatial randomness (CSR). In this model, the points of a spatial pattern are stochastically independent. The nature of the phenomenon under study or a casual glance at a plot of the data usually makes it obvious when CSR is not a realistic option.

The nature or behaviour of a point pattern may be thought of as comprising two components, trend and dependence, or interaction between the points of the patterns. The simplest manifestation of such interaction consists of either attraction (aggregation or clustering) or repulsion (regularity) in the pattern. A useful step in analysing a point pattern is to apply graphical tools which reveal information as to the nature of the interaction. A widely used tool for exploring the nature of interaction is Ripley's K-function (Ripley, 1976, 1977; Cressie, 1993; Diggle, 2003).

The basic idea in interpreting the K-function is that a constant intensity Poisson process (a process exhibiting CSR) has a K-function equal to $K(r) = \pi r^2$. If there is attraction (with impact at distance r, the bounding radius of the spatial domain), then $K(r)$ is larger than it would be under CSR. Conversely, if there is repulsion, then $K(r)$ is smaller than it would be under CSR.

2.2.1. Spatial models

2.2.1.1. Poisson processes. The homogeneous Poisson process is the simplest point process that represents no underlying process, corresponding in our case to complete randomness in wildfire distribution. The K-function of the homogenous Poisson process is defined as:

$$K(r) = \frac{1}{A}\sum_{i=1}\sum_{j\neq 1}\frac{\omega_{ij}I(d_{ij}\leq r)}{\lambda^2}$$

In this function, A denotes the area of the plot; λ is wildfire density, ω_{ij} is an edge correction term, d_{ij} represents the distance between two points, and I is an index function where $I = 1$, if $d_{ij} \leq r$, and $I = 0$ otherwise (Ripley, 1976). Wildfire density, λ, is the parameter to be estimated in this model.

The inhomogeneous Poisson process can be used to model heterogeneous association in wildfires. In this model, relationships between density and heterogeneity are included via a spatially heterogeneous intensity function, $\lambda(s)$ (Diggle, 2003; Illian et al., 2008). The K-function of the inhomogeneous Poisson process is defined as

$$K_{\mathrm{inh}}(r) = \frac{1}{A}\sum_{i=1}\sum_{j\neq 1}\frac{\omega_{ij}I(d_{ij}\leq r)}{\lambda(s_i)\lambda(s_j)}$$

where A, λ, ω_{ij}, d_{ij}, and I are the same as in the homogeneous Poisson process; and $\lambda(s_i)$ and $\lambda(s_j)$ are the values of the intensity function at points s_i and s_j, respectively (Diggle, 2003; Illian et al., 2008).

Specifically, the intensity function, $\lambda(s)$, is modelled as a log-polynomial regression:

$$\lambda(s) = \exp\left(\beta^T X(s)\right)$$

where $X(s)$ is a vector of variables and β^T is a vector of regression parameters. Two different types of log-polynomial regressions were used in this study: log-linear regressions with covariates and log-quadratic regressions with the coordinates of the wildfire.

2.2.1.2. *Thomas processes.* The homogeneous Thomas process is a particular class of Poisson cluster process and can be used to model a series of clustered patterns (Diggle, 2003; Illian et al., 2008). This model describes processes of dispersal, in which 'offspring' are limited to aggregate around their 'parent'. Therefore, it models the effect of dispersal limitation. The homogeneous Thomas process is modelled by two steps. First, locations of parents are generated by a homogeneous Poisson process with a density, j. Second, a group of offspring are produced around each parent. Their locations are assumed to be independent of one another and isotropically distributed around each parent with a Gaussian dispersal kernel, $N(0,r)$. The number of offspring is determined by a Poisson distribution with mean being l (Moller and Waagepetersen, 2004; Baddeley and Turner, 2005).

The K-function of the homogenous Thomas process is given by:

$$K(r) = \pi r^2 + \frac{1-e^{(-r^2/4\sigma^2)}}{\kappa}$$

where r is distance, κ represents the intensity of parents in a Poisson distribution and σ is standard deviation of distance from offspring to the parent (Diggle, 2003; Illian et al., 2008). Mean number of offspring per parent in a Poisson distribution, l, can be inferred from estimated intensity λ and κ.

The inhomogeneous Thomas process is used to evaluate the joint effects of covariates (Diggle, 2003; Illian et al., 2008). This model is the same as a homogeneous Thomas process, except that the number of offspring per parent, I, is no longer constant and must be estimated by a spatially heterogeneous intensity function. As with the inhomogeneous Poisson process above, intensity functions were modelled by means of log-polynomial regressions.

2.2.1.3. *Area-interaction processes*. The homogeneous area-interaction process (Widom and Rowlinson, 1970; Baddeley and van Lieshout, 1995) with disc radius r, intensity parameter κ and interaction parameter γ is a point process with probability density:

$$f(s_1,\ldots,s_n) = \alpha\kappa^{n(s)}\gamma^{(-A(s))}$$

where $s_1,\ldots s_n$ represent the points of the pattern, $n(s)$ is the number of points in the pattern, and $A(s)$ is the area of the region formed by the union of discs of radius r centred at the points $s_1,\ldots s_n$. Here, α is a normalizing constant.

The interaction parameter γ can be any positive number. If $\gamma = 1$ then the model reduces to a Poisson process with intensity κ. If $\gamma < 1$ then the process is regular, while if $\gamma > 1$ the process is clustered. Thus, an area interaction process can be used to model either clustered or regular point patterns. Two points interact if the distance between them is less than $2r$.

Here, we parameterized the model in a different form (Baddeley and Turner, 2005), which is easier to interpret. In canonical scale-free form, the probability density is rewritten as

$$f(s_1,\ldots,s_n) = \alpha\beta^{n(s)}\eta^{(-C(s))}$$

where β is the new intensity parameter, η is the new interaction parameter, and $C(s) = B(s) - n(s)$ is the interaction potential. Here

$$B(s) = A(s)/\left(\pi\cdot r^2\right)$$

is the normalized area (so that the discs have unit area). In this formula, each isolated point of the pattern contributes a factor β to the probability density (so the first order trend is β). The quantity $C(s)$ is a true interaction potential, in the sense that $C(s) = 0$ if the point pattern x does not contain any points that lie close together (closer than $2r$ units apart).

The old parameters κ and γ of the standard form are related to the new parameters β and η, of the canonical scale-free form, by

$$\beta = \kappa\gamma^{(-\pi\cdot r^2)} = \kappa/\eta$$

and

$$\eta = \gamma^{(-\pi\cdot r^2)}$$

provided κ and γ are positive and finite (Baddeley and Turner, 2005).

In the canonical scale-free form, the parameter η can take any non-negative value. The value $\eta = 1$ again corresponds to a Poisson process, with intensity β. If $\eta < 1$ then $\gamma > 1$ for any value of r, and the process is clustered. If $\eta > 1$ then $\gamma < 1$, and the process is regular. The value $\eta = 0$ corresponds to a hard core process with hard core radius r (interaction distance $2r$).

The inhomogeneous area interaction process is similar, except that the contribution of each individual point s_i is a function $\beta(s_i)$ of location, rather than a constant beta.

2.2.2. *Statistical inference*

To fit Poisson cluster models, such as the Thomas model, and given a user-specified maximum distance h_k, the model parameters can be estimated by minimizing a 'discrepancy measure' of the empirical K-function $\widehat{K}(h)$ and the expected value $K(r)$ (Diggle, 1983):

$$\int_0^{h_k}\left(\widehat{K}(h)^{0.25} - K(r)^{0.25}\right)^2 \mathrm{d}h$$

And for the inhomogeneous case:

$$\int_0^{h_k}\left(\widehat{K}_{\mathrm{inhom}}(h)^{0.25} - K(r)^{0.25}\right)^2 \mathrm{d}h$$

These are known as minimum contrast procedures.

Because the choice of h_k is arbitrary, Diggle (2003) recommended that h_k should be considerably smaller than the dimensions of the observation plot.

When spatial interactions exist, and a likelihood is obtained in closed form, the model parameters can then be estimated via the maximum pseudo-likelihood method. Goodness-of-fit of the models was evaluated using Akaike's information criterion (AIC) and Monte Carlo simulations. The AIC was used to select the best-fit model for each set of wildfires and was calculated using the following formula:

$$\mathrm{AIC} = n\ln(R) + 2k$$

where n is the number of observations, R is the sum of residual squares and k is the number of parameters (Moller and Waagepetersen, 2004; Shen et al., 2009). Number of parameters ranged from 1 to 12 for the various models. The AIC was used because in this study parameters were not estimated using standard maximum likelihood methods. Goodness-of-fit was further evaluated by means of Monte Carlo simulations, which were used to generate 95% confidence intervals of the $K(r)$ for different models.

To fit an area-interaction process we made use of maximum pseudo-likelihood based-methods (Baddeley and Turner, 2005). The pseudo-likelihood estimation approach provides an alternative to likelihood methods when the normalizing constant is no longer to be determined. This pseudo-likelihood function is based on the conditional intensity of a point s_i given realization in a bounded region A. For

Table 1
Distribution of wildfires occurring in Catalonia by year and cause.

Number of fires	Cause 1 (naturals)	Cause 2 (negligence)	Cause 3 (intentional)	Cause 4 (unknown)	All causes
2004	65	336	103	59	563
2005	115	525	151	102	893
2006	98	297	155	78	628
2007	52	308	122	96	578
2008	38	227	94	62	421
All years	368	1693	625	397	3083

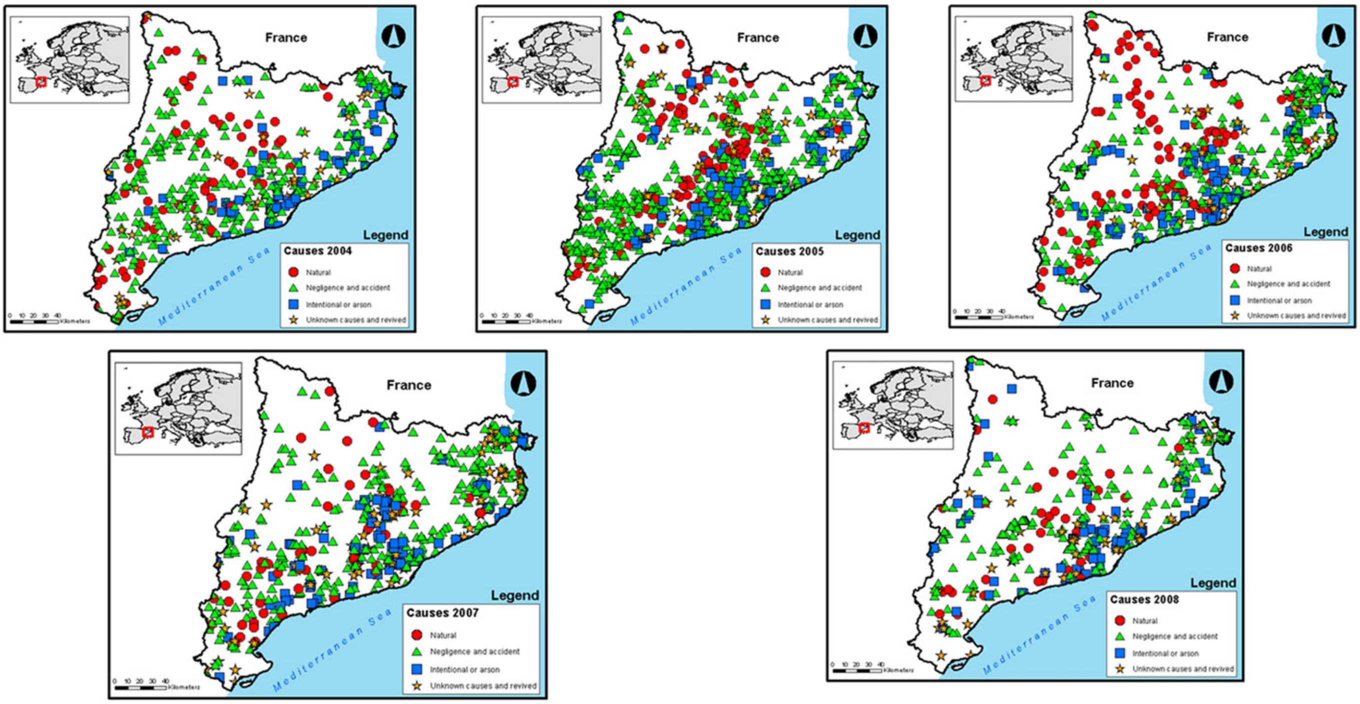


Fig. 1. Distribution of wildfires occurring in Catalonia by year and cause (natural; negligence or accident; intentional or arson; unknown causes and revived).

pairwise interaction processes, the conditional intensity (Papangelou, 1974; Daley and Vere-Jones, 2003) of ϕ at a location s_i may be loosely interpreted as giving the conditional probability that ϕ has a point at x given that the rest of the process coincides with ϕ. For $s_i \notin \phi$ and $f(\phi) > 0$, the conditional intensity equals:

$$\lambda(s_i;\phi) = \frac{f(\phi \cup \{s_i\})}{f(\phi)}$$

while for $s_i \in \phi$ we have $\lambda(s_i;\phi) = f(\phi)/f(\phi - \{s_i\})$. If η (i.e. pair potential function) is bounded then for any finite point configuration ϕ with $f(\phi)$, the pseudo-likelihood of pairwise interaction processes is defined by (Besag, 1974; Jensen and Moller, 1991),

$$\mathrm{PL}(\phi;\theta) = \exp\left(-\int_A \lambda(s_i;\phi)\mathrm{d}s\right)\prod_{i=1}^{n}\lambda(s_i;\phi)$$

Usually, this is re-cast in terms of its logarithm. Maximization of the PL function with respect to the parameter set yields the maximum pseudo-likelihood estimators.

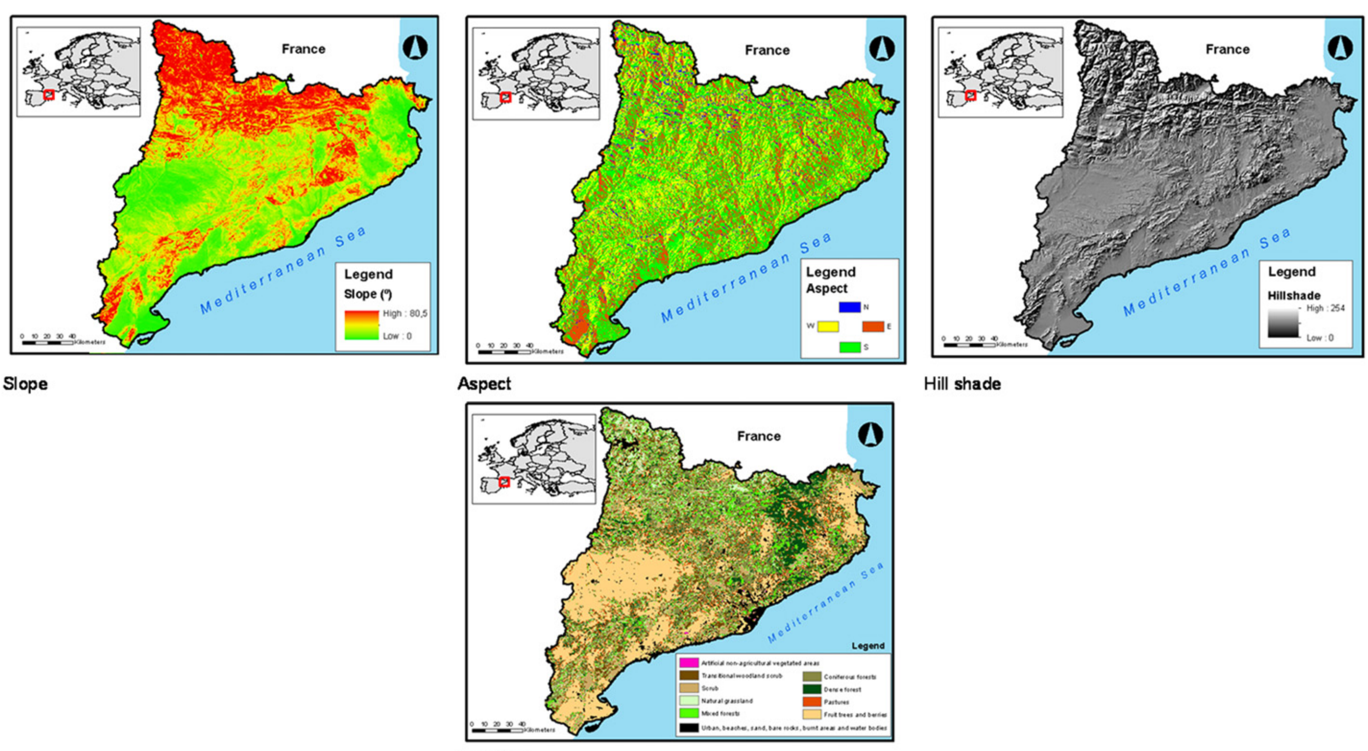


Fig. 2. Spatial covariates, both continuous (slope, aspect and hill shade) and categorical (land use).

Table 2
The twenty models with best fit to the observed spatial distribution of the wildfires.

Year	2004				2005				2006				2007				2008			
Cause	C1	C2	C3	C4	C1	C2	C3	C4	C1	C2	C3	C4	C1	C2	C3	C4	C1	C2	C3	C4
Poisson					×								×	×			×			
Thomas	×				×				×								×			
Area-Interaction	×	×	×	×	×			×		×	×		×	×			×		×	

2.3. Analysis of spatial segregation

With the double aim to assess our first objective and to provide additional evidence of the deviation of our data from the null hypothesis of CSR, we follow Diggle et al. (2005) and investigate the occurrence of spatial segregation. Spatial segregation occurs if, within a region of interest, some types of points dominate in some sub regions, and this dominance is determined by circumstances other than random.

Data are now represented as a set of multinomial outcomes Y_{it}, $i = 1,\ldots,n$, $t = 2004, \ldots, 2008$ ($n = 3083$ wildfires from 2004 to 2008; 563 in the year 2004; 893 in 2005; 628 in 2006; 578 in 2007; and 421 in the year 2008), where, for each of $k = 1,\ldots,m$ ($m = 4$, 1 = natural causes; 2 = negligence and accidents; 3 = intentional or arson; 4 = unknown causes and revived), and year t, the outcome $Y_{it} = k$ denotes the occurrence of a wildfire for the cause k at the location x_i the year t, and the corresponding multinomial cell probabilities are the cause-specific probabilities $p_k(x)$,

$$p_k(x) = \frac{\lambda_k(x)}{\sum_{j=1}^{m} \lambda_j(x)}$$

where $\lambda_k(x)$ denotes the intensity function of the independent Poisson process corresponding to cause k.

The cause-specific probabilities, $p_k(x)$, were estimated through a multivariate adaptation of a kernel smoothing. In particular, the cause-specific probability surfaces were estimated by means of a kernel regression estimator,

$$\widehat{p}(x) = \sum_{i=1}^{n} w_{ik}(x) I(Y_i = k)$$

where $I(\cdot)$ was an indicator function; and, for each $k = 1,2,\ldots,m$,

$$w_{ik}(x) = \frac{w_k(x - x_i)}{\sum_{j=1}^{n} w_k(x - x_j)}$$

$w_k(\cdot)$ was the kernel function with bandwith $h_k > 0$,

$$w_k(x) = \frac{e^{(-\|x\|^2/2)}}{h_k^2}$$

where we used the Gaussian kernel as the standardized form of the kernel function in the numerator; and $\|x\|$ denotes the Euclidean distance of the point x to the origin.

Estimation of the cause-specific probabilities was done by means of cross-validated log-likelihood (Diggle et al., 2005). For the testing of both, the null hypotheses of no spatial variation in the probability surfaces between wildfire causes (i.e. no spatial segregation); and of no change over time of the cause-specific probability surfaces (i.e. no temporal changes in spatial segregation) we used Monte Carlo sampling (Diggle et al., 2005).

The analyses were carried out using the R freeware statistical package (versions 2.12.1 and 2.14.2) (R Development Core Team, 2010); and the *Spatstat* (Baddeley and Turner, 2005); *splancs* (Rowlingson and Diggle, 1993) and *spatialkernel* (Zheng et al., 2012) packages. The free environment *gvSIG (version 1.10)* was used for representing maps.

3. Results

Table 1 and Fig. 1 show the distribution of the 3083 wildfires occurring in Catalonia by year (from 2004 to 2008) and cause (natural causes; negligence and accidents; intentional or arson; and unknown causes and revived).

Table 3
Results of estimating the twenty models with best fit to the observed spatial distribution of the wildfires.

Thomas							
	β_0	(Aspect)β_1	(Hill shade)β_2	(Slope)β_3	(Land Use)β_4	κ	σ
2004							
Cause 1	4.2783	−0.0004	0.0026	0.0235	0.0735	53.3747	0.0432
2005							
Cause 1	5.4771	−4.848e-06	−0.0011	0.0119	0.0463	18.4070	0.0549
2006							
Cause 1	4.7536	0.0001	0.0013	0.0025	0.0879	60.5309	0.0338
2008							
Cause 1	4.7579	−0.0009	−0.0003	0.0192	0.0411	45.2217	0.0543
Areal-interaction							
	β_0	(Aspect) β_1	(Hill shade) β_2	(Slope) β_3	(Land use) β_4	η	Interaction
2004							
Cause 1	4.2446	−0.0003	0.0024	0.0145	0.0571	31.563	3.4520
Cause 2	7.2342	−0.0014	−0.0002	−0.0068	−0.0135	7.0907	1.9588
Cause 3	5.2597	−0.0024	0.0009	0.0251	0.0018	254.80	5.5405
Cause 4	6.1766	−0.0028	−0.0015	0.0011	0.0128	4.0229	1.3920
2005							
Cause 1	5.5155	−0.0002	−0.0011	0.0249	0.0456	0.8201	−0.1983
Cause 4	5.6711	−0.0006	0.0008	0.0073	−0.0671	70.431	4.2546
2006							
Cause 2	7.2418	−0.0023	−0.0006	0.0168	0.0025	8.3818	2.1261
Cause 3	5.5181	−0.0011	−0.0001	−0.0002	−0.0331	313.024	5.7463
2007							
Cause 1	4.8859	−0.0007	−0.0005	0.0330	−0.0041	48.078	3.8728
Cause 2	6.9511	−0.0018	0.0009	0.0010	0.0230	6.0967	1.8077
2008							
Cause 1	4.6428	−0.0008	−0.0007	0.0188	0.0336	32.852	3.4920
Cause 3	5.7116	−0.0027	0.0007	0.0038	−0.0785	201.55	5.3060

The category of negligence and accidents was the most frequent cause (54.91%), followed by intentional or arson (20.27%), unknown (12.88%) and natural causes (11.94%). Note also that after a dramatic increase in the number of wildfires from 2004 to 2005 (58.61%), there was then a decrease (29.68% from 2005 to 2006; 7.62% from 2006 to 2007; and 27.16% from 2007 to 2008). In fact, 2008 was the year with the lowest number of wildfires during the study period. Note that the behaviour of each of the causes by year was more or less the same, with a slight difference in the case of unknown and revived, where the year with the lowest number of wildfires was 2004. The spatial distribution of the wildfires was very similar for all causes, perhaps with the exception of natural causes, with very

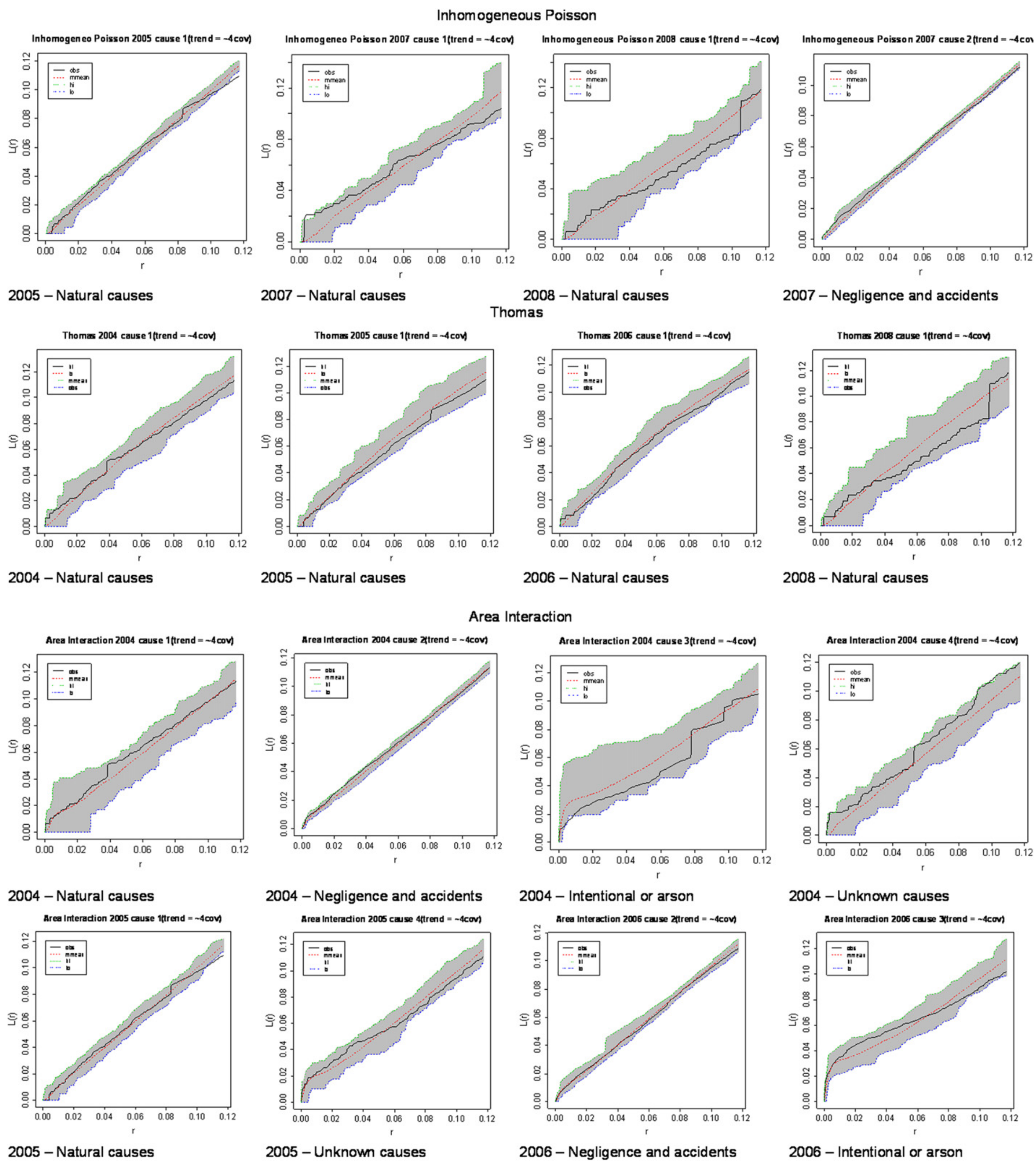


Fig. 3. Simulated envelopes of the twenty models with best fit to the observed spatial distribution of the wildfires.

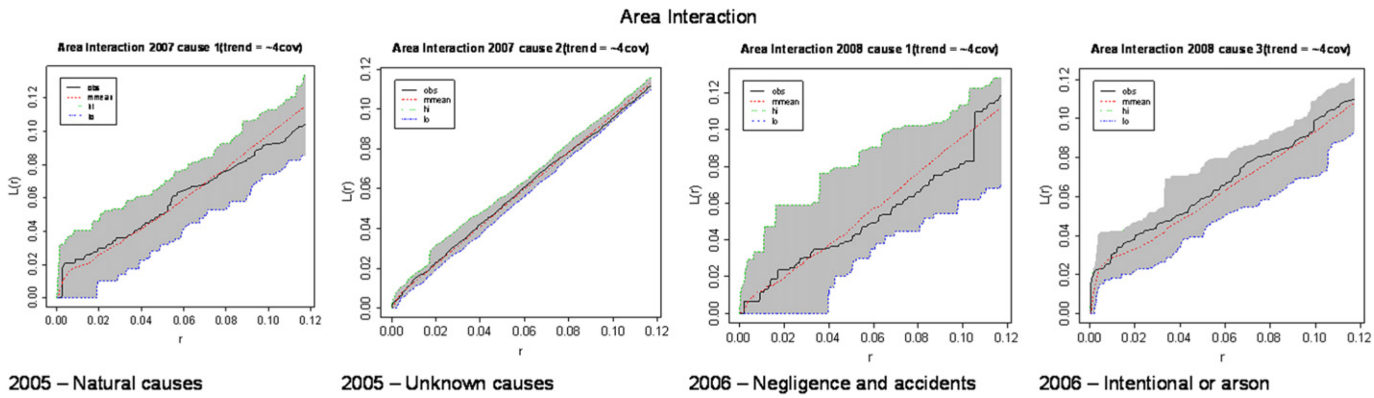


Fig. 3. (*continued*).

Table 4
AIC of best Poisson and area-interaction models (common years and causes).

	AIC	
	Poisson	Area-interaction
2005 Cause 1	−1023.863	−1025.464
2007 Cause 1	−386.7365	−386.2792
2007 Cause 2	−3512.036	−735.2788
2008 Cause 1	−255.3777	−254.4718

few fires in coastal and urban areas (which are located mainly on the coasts). Note also, that intentional and unknown causes behaved more like each other than like other causes.

Although it would be theoretically possible that the use of topographic variables would produce high collinearity, this problem did not occur in practice. Correlation between hill shade and aspect was equal to 0.402 ($p < 0.001$); between hill shade and slope −0.360 ($p < 0.001$); and −0.025 ($p = 0.158$), between aspect and slope. Nevertheless, we repeated all analyses using: i) slope and aspect separately (without hill shade); and ii) hill shade alone (without slope and aspect). However, the results vary very little. In neither model parameter estimators suffer a significant change and, in any case, vary the parameters of statistical significance.

To model the behaviour of the wildfires we have used three different models for each year and cause. Initially, we individually entered each covariate (slope, aspect, hill shade and land use, see Fig. 2) into each of the models. However, due to the large number of resulting models for all possible combinations, we decided to focus the study solely on the four covariates together, as this already covered individual results and gave other general information.

Discarding the homogeneous models, since intensity is always clearly variable in the case of modelling wildfire, we started with the simplest model, the inhomogeneous Poisson process. We followed this with a cluster process model, specifically the Thomas process, and finished with the area-interaction point process. In total sixty models were fitted to the data. Among them, and according to the fit of the model to the observed spatial distribution of the wildfires, we selected the twenty best models (see Table 2). It is interesting to note that the Thomas process was only well modelled for natural causes for practically all years (except 2007). Note that the area-interaction model shows a clear improvement over the other two models. It properly adjusted most cases including, unlike the other two, the cause intentional or arson. The results of estimating the twenty best models are shown in Table 3 and the simulated envelopes of the fit are shown in Fig. 3.

Models were selected using a second filtration taking into account the AIC. As is known, this criterion measures the relative goodness-of-fit of the model in question and is therefore an effective tool for comparing models. The lower the AIC, the better the model (in terms of goodness-of-fit). AIC can be computed in terms of close likelihoods in which the number of parameters playing a role in the specific model are considered. For some cluster models, such as the Thomas one, the close likelihood can not be computed and thus AIC could also be calculated in an approximate way, making the comparison with Poisson or Area-Interaction models unfair. In this sense, we only used AIC criteria for these latter two models (see Table 4). In these cases, there was generally little difference between the two models analysed, except for the cause negligence and accidents in the year 2007, where the Poisson model had the lowest AIC.

The analysis of spatial segregation is shown in Table 5. All four cause-specific probabilities show wide spatial variation for all years. Using 999 simulated random relabellings of the wildfire causes among all case in the Monte Carlo tests for both, spatial segregation and the temporal changes of the spatial segregation, we rejected the null hypotheses each one of the years and causes, respectively.

Table 5
Analysis of the spatial segregation.

	2004	2005	2006	2007	2008	Monte Carlo test for temporal changes of spatial segregation
Cause-specific probabilities[a]						
Natural causes	0.44982–5.353e–07	0.44781–1.262e–07	0.27130–0.00245	0.42920–0.00523	0.34484–0.05102	0.034
Negligence and accidents	0.52385–2.719e–05	0.91912–0.01334	0.89029–0.00814	0.26738–0.00135	0.37881–0.00833	0.002
Intentional or arson	0.88786–0.31920	0.69422–0.00056	0.81789–0.01535	0.43960–0.00011	0.81901–0.17470	0.021
Unknown causes and revived	0.47607–9.673e–06	0.46596–0.00674	0.55427–0.00300	0.88825–0.35331	0.39663–0.01364	0.044
Monte Carlo test for spatial segregation	0.002	0.012	0.003	0.002	0.020	

[a] Maximum–minimum.

Note, that there was a great variability in the estimated effects of the variables on the intensities for all causes, years and spatial point process models finally fitted (see Table 4). As a result, conditional intensities should vary accordingly.

Finally, we computed the conditional intensity of all fitted spatial point process models in Table 2, evaluated at each spatial location used for model fitting. Our interest was to use these intensities to build maps of wildfire risks, by year and cause of

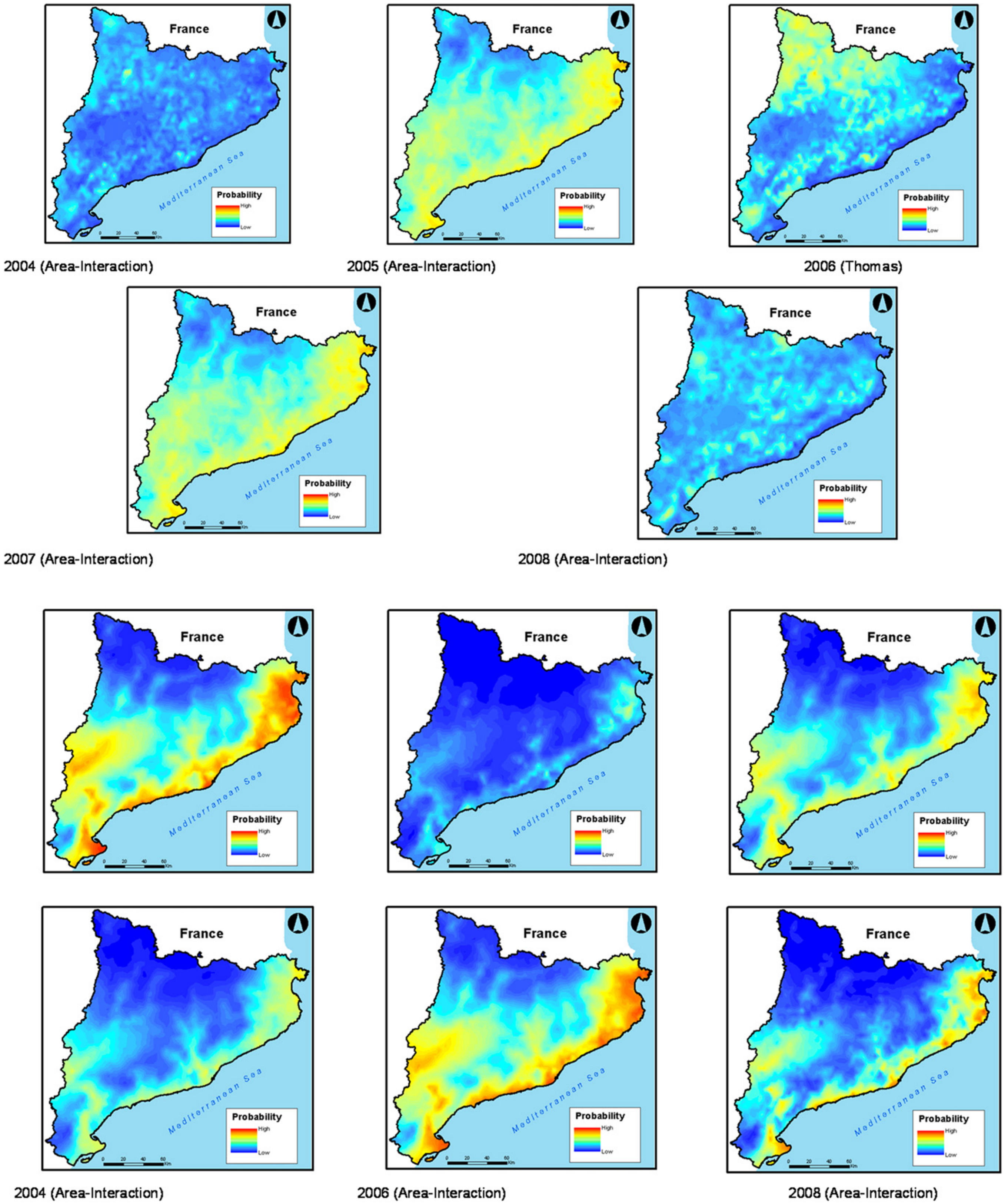
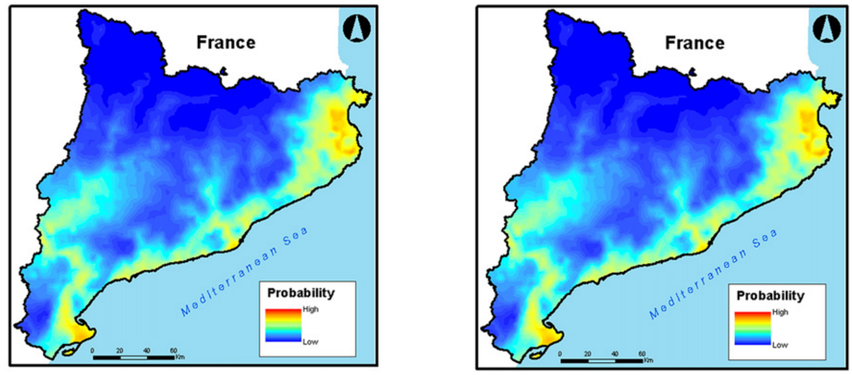


Fig. 4. Conditional intensity of the fitted spatial point process models. Natural causes.



2004 (Area-Interaction)

2005 (Area-Interaction)

Fig. 4. (*continued*).

ignition (Fig. 4), in order to provide a tool for prevention and management of vulnerability levels. Here we show only those maps corresponding to the best models (in terms of goodness-of-fit), i.e. area-interaction with two exceptions, natural causes in 2006, where the only model was the Thomas process (see Table 2), and negligence and accidents in 2007, built using the Poisson model (see Table 4) (results not shown can be requested from the authors).

With some exceptions (unknown causes and revived in 2004 and 2005, and natural causes in 2005 and 2007) the risks for each cause varied over the years. Note that, generally speaking, risks for natural causes in particular and, to a lesser extent, unknown causes and revived, were not very high, whereas for negligence and accidents (2004 and 2007) and intentional or arson (2006 and 2008) risks reached figures close to the maximum in some areas, mainly coastal and most urbanized areas.

Wildfires attributed to natural causes are where we find more variability: each year is completely different from the rest. This is due to the fact that these fires are basically caused by dry lightning storms (no rain); this phenomenon can occur anywhere: mountains, sea, etc. With anthropogenic causes, on the other hand, we see a clear pattern in the coastal areas (especially Barcelona and the Costa Brava), where fires occur in July and August, when more tourists can be found in these places.

4. Discussion and conclusions

This study has three main findings. Firstly, the extent of clustering in wildfires differed across marks, i.e. years and cause of wildfire ignitions. Secondly, covariates such as land use, slope, aspect and hill shade influenced trends in the intensity of wildfire locations. Finally, we have built maps of wildfire risks, by year and cause of ignition, from the estimated models. These could prove very useful as tools in preventing and managing vulnerability levels.

As regards the first finding, we conclude that, despite the variability found among marks, especially over time, the model that best fits the behaviour of fires for most years and causes is the area-interaction point process model. This result is interesting because it will allow us to expand the study to space–time modelling and will lead to improved prediction of fire risk.

Regarding the second finding, the analysis should be completed using other covariates such as fuel (present at the location of fire), flammability, proximity of the wildfire to roads and towns, and economic factors such as land prices. On the other hand, it would also be interesting to introduce some meteorological covariates such as temperature, precipitation and wind.

Note that we have restricted our attention to inhomogeneous spatial models where we have fixed the temporal scale and modelled only the spatial component. A necessary and further improvement of our modelling efforts might incorporate time into the model itself, considering thus space–time point process models. One good thing about this latter approach is that we could model and evaluate the corresponding space–time interaction, something that we have not considered in the present approach. We note that some approaches have already been considered in this respect, but they consider independent spatial replication in time, and this is not realistic. In the context of space–time modelling, one useful approach is to try and model the space–time intensity function as an additive or multiplicative form of the spatial and temporal intensities, and then adding a space–time residual component (for instance Diggle et al., 2005b). This would probably provide more insight into the problem of modelling wildfires.

With respect to the third finding, we constructed maps that could: i) help authorities to develop more reliable plans for forest fire prevention; ii) predict areas of forest fire risk more efficiently; iii) design awareness campaigns more efficiently; and iv) reduce the budget designed of fire management. In fact, if we understand prevention as prediction of scenarios on where and when forest fires could be stronger in critical moments, our maps could provide the necessary land-use planning measures, both in terms of preventive management practices and stakeholders' responsibility in managing fires; and identify priority actions for each component of the fire management system throughout the year.

As we said, hill shade was defined as a function of slope and aspect (along with angle of illumination). In fact, the correlation between hill shade-aspect and hill shade-slope were actually quite high (about 0.4 in each case). However, it seems that there is some property of the stochastic models which makes them insensitive to inter-correlation.[1] In fact, when we repeated all analyses using: i) slope and aspect separately (without hill shade); and ii) hill shade alone (without slope and aspect) the results vary very little.

The methodology used has allowed us to quantify and assess possible spatial relationships between the distribution of risk of ignition and causes. These results may be useful in fire management decision-making and planning. We believe the methods shown in this paper may contribute to the prevention and management of wildfires, which are not random in space or time, as we have shown here.

[1] We thank an anonymous reviewer for pointing out this point.

Conflicts of interest

There are no conflicts of interest for any of the authors. All authors disclose any actual or potential conflict of interest including any financial, personal or other relationships with other people or organizations within three years of beginning the submitted work that could inappropriately influence, or be perceived to influence, their work.

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